

DPCN Assignment 1

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1 Team Name

Team Edge Runners : Arijeet Paul, Shreyash Lohare, Dev Patel.

2 GitHub Link to Code

All code, figures and data are at https://github.com/shreyash-lohare/dpcn_a1_opinion_network.

3 Dataset Documentation

The questionnaire contains 60 items in four blocks of 15 — technology (T01–T15), education (E01–E15), society and ethics (S01–S15) and environment (V01–V15) — answered by 96 respondents. Five ordered categories are encoded symmetrically as $SD = -2$ through $SA = +2$, which represents the ordering without asserting equal spacing; a sixth option, No Comments (NC), is treated as missing. Internal consistency varies sharply by block: Cronbach’s α is 0.87 for environment and 0.82 for society but only 0.65 for technology and 0.58 for education.

3.1 The response ceiling

The most consequential property of this dataset is its ceiling. Figure 1 shows 79.8% of valid answers on the agree side, 43.1% Strongly Agree, and only 7.0% disagreeing. The grand mean is **+1.14** on the $[-2, +2]$ scale. Two consequences drive every later decision: item means barely discriminate, so the signal is in which statements attract disagreement; and two respondents look similar under any raw similarity measure simply by both being agreeable.

The ceiling is not uniform. Environment ($\mu = 4.39$ on the 1–5 scale, 89.5% agree) and society ($\mu = 4.28$) are near consensus, while technology ($\mu = 3.99$) and education ($\mu = 3.91$, 13.8% disagree) retain real disagreement — and those two blocks also have the weakest internal consistency, consistent with their items tapping several distinct opinions rather than one settled disposition.

Education and technology contribute most of the twelve most contested items; environment contributes

none. Compulsory attendance (E03, 70.1% disagreeing) is the only statement the cohort actively rejects, yet the same cohort agrees at 94.3% that curricula should be updated more often — the objection is to assessment and teaching format, not to the institution. Technology shows the same pattern: AI-assisted diagnosis and autonomous vehicles are its least popular items, while teaching AI and robots in hazardous work both score above 79%, so the objection tracks delegating consequential judgment rather than automation itself.

3.2 Missing data and how it was recovered

Of 96 records, 85 are complete, 5 are entirely empty, and 6 stop partway. The five empty records are dropped, leaving 91 respondents; none is dropped merely for being incomplete. That leaves 205 structural blanks plus 37 NC cells, **242** in total. Dropout is monotone — a truncated tail, not scattered gaps — so later blocks rest on a slightly smaller base.

NC is not Neutral. Recoding a withheld answer as Neutral would assert a response nobody gave: a simple model fitted on the clean responses predicts 3.92 at the NC cells, not 3, so recoding would impose a systematic downward shift. NC use also shows no pattern by item or by respondent, so there is no basis for assigning it any fixed category.

Recovery by ordinal regression. Each missing cell is predicted from a respondent’s other 59 answers by a regularised ordinal regression fitted per item, which respects the order of the five response categories without assuming equal spacing between them (the full method is in Section 4). We validate this by artificial masking under two mechanisms — scattered cells and a structural tail, mimicking NC and drop-out respectively — and compare against a simple modal-guess baseline (Table 1).

One caution carries forward: different ways of handling NC give correlation matrices that agree closely ($r > 0.94$) but community assignments that can disagree sharply (AMI as low as 0.337). A stable correlation matrix does not guarantee a stable partition of it — a point Section 6.2 returns to.

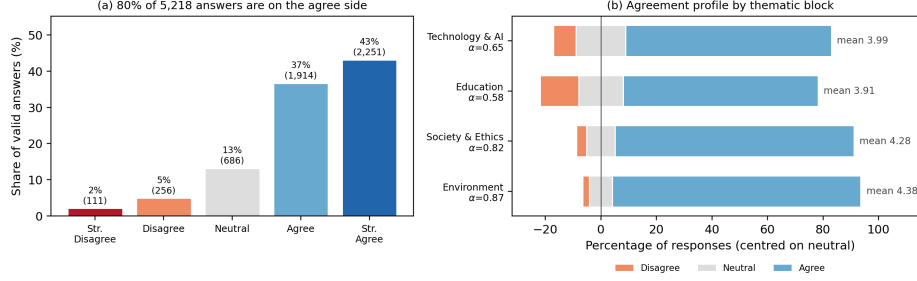


Figure 1: Response ceiling and block-level agreement. (a) All 5,218 valid answers: nearly four in five are on the agree side. (b) Agreement profile per block with block mean and Cronbach’s α . Environment and society are near consensus; education carries the most disagreement and the weakest consistency.

Table 1: Artificial-masking validation of the ordinal-regression imputer. The model beats a modal baseline under both mechanisms, by a wider margin where more co-answered items survive.

Masking	Ordinal model		Mode baseline	
	Acc.	MAE	Acc.	MAE
Scattered (NC-like)	.539	.534	.486	.626
Suffix (structural)	.498	.612	.482	.655

3.3 Row-centring

Imputation yields a complete 91×60 matrix, but not one that can be correlated directly: with a grand mean of +1.14, a respondent who agrees with everything raises all 60 entries together, which reads as association between every pair of items at once. We replace each entry by its deviation from that respondent’s own mean,

$$x'_{i,j} = x_{i,j} - \bar{x}_{i\cdot}, \quad \bar{x}_{i\cdot} = \frac{1}{60} \sum_k x_{i,k}. \quad (1)$$

Centring is applied strictly after imputation and strictly before correlation. Note that rank correlation does not remove this bias automatically: Spearman is invariant to a monotone transform of a column, not to an additive shift that differs by row. Graph 3 (Section 5.3) measures how much it matters.

4 Pipeline Followed

4.1 What the network is

Vertices are the 60 questionnaire items, and edges are signed correlations between items, computed on the row-centred matrix and retained with their sign. Edges are weighted by default and binarised only where a metric requires it (path lengths, triad counts).

Items were chosen over respondents deliberately: an item network is thresholded directly on a measured

quantity and every vertex carries an interpretable label, whereas a respondent network has no directly measured edge and would have to be constructed by an imposed neighbourhood rule. Signs are kept because a negative edge — two questions dividing the cohort in opposite directions — is a stronger statement than mere association; Graph 8 (Section 5.8) shows this sign pattern separating topic blocks cleanly once acquiescence is removed.

4.2 Step by step

1. Parse and encode. Identify the 60 item columns, map the five categories to $\{-2, \dots, +2\}$ and NC to NaN. Yields 96×60 with 542 missing cells.
2. Filter respondents. Drop only the five entirely empty records. 91×60 , 242 missing cells.
3. Impute (Section 3.2). One regularised proportional-odds ordinal logistic regression per item, predicting it from the other 59.
4. Row-centre (Section 3.3). Subtract each respondent’s own mean across all 60 items — strictly after imputation, strictly before correlation.
5. Correlate. Pearson correlation over the 60 columns, giving a 60×60 signed matrix with $\binom{60}{2} = 1,770$ distinct off-diagonal entries.
6. Establish the noise floor (Section 5.4). Re-run steps 4–5 on 200 independently column-permuted copies of the matrix from step 3.
7. Denoise (Section 5.5). Keep only the eigenvalue modes exceeding the Marchenko–Pastur upper edge λ_+ , rebuild the matrix from those modes, and restore the unit diagonal.
8. Threshold (Section 5.7). Sweep all 46 cutoffs from 0.05 to 0.50 and select by a stated rule rather than by inspecting the result. Selects $|r| > 0.22$.
9. Measure (Section 5). Community detection (Graphs 6 and 9) and topic-block connectivity (Graph 8) — each compared against an explicit null.

5 Analysis and Visualisations

This section reports the nine analyses. Each states what the figure shows, what was found, and what follows from it.

A note on respondent counts. All figures below report $n = 91$: every non-empty respondent is retained and the missing cells are recovered by ordinal regression (Section 3.2). An alternative pipeline that instead drops respondents with more than 12 missing items, leaving 86, is examined in Section 6.2.

5.1 Graph 1: Missingness map and response distribution

Figure 2 shows every cell of the raw 96×60 matrix as missing or present, respondents sorted by missing count, beside the distribution of per-respondent means.

Findings. The 542 missing cells (505 structural blanks plus 37 NC) are extremely unevenly distributed: five entirely empty respondents account for 300 of them, and the remaining 242 spread thinly across the other 91 — a rate of 4.4%. Below the cutoff the map is almost clean, with two visible tails from respondents who stopped early. The right panel locates the grand mean at 1.14, with only one respondent below zero.

Implication. This justifies the retention policy directly: all but five respondents answered at least 80% of items, so there is no population of badly incomplete respondents whose removal would simplify anything, and a count-based exclusion rule would discard usable respondents to remove very few cells. The right panel is the first direct evidence of the acquiescence problem that row-centring corrects.

5.2 Graph 2: Item variance ranking

Figure 3 ranks all 60 items by standard deviation on the uncentred matrix. Centring is deliberately not applied here: it changes each item’s variance and would answer a different question (spread after removing each respondent’s baseline) than the one asked (how divided the cohort is on the statement as posed).

Findings. Item SD ranges from 0.50 (E15, near-unanimous) to 1.18 (E04, online learning). The five most contested are E04, E03, E02, T08, T07. By block mean SD: technology 0.92, education 0.88, society 0.79, environment 0.72.

Implication. Technology has the highest average variance, but the sharpest disagreements are educational — three of the five most divided items are E02–E04, all about assessment and teaching format — and environment contributes none of the top ten. This is a property of the instrument, not of preprocessing: the

questionnaire contains one settled domain, one moderately settled, and two live ones. Consensus items carry little information for a correlation network, since an item everyone endorses cannot covary with anything, so the signal is necessarily concentrated in the contested tail.

5.3 Graph 3: The effect of row-centring

Figure 4 overlays all $\binom{60}{2} = 1,770$ pairwise item correlations before and after row-centring, on identical bin edges.

Findings. The raw distribution is displaced positive (mean and median both 0.151), with almost no pair appreciably negative. After centring it sits on zero (mean -0.004 , median -0.008) with positive skew 0.221, leaving a heavier right tail of genuinely co-varying pairs. At $|r| \geq 0.30$ the surviving pairs fall from **327** to **42** — a reduction of 87%.

Implication. This converts the centring argument from an assertion into a measurement: seven eighths of the apparent association in the raw matrix is shared response style. Built on raw scores the network would have been dense, almost entirely positive, and would have supported confident but spurious conclusions. The emergence of negative correlations after centring — invisible in the raw distribution — is also what makes the sign pattern in Graph 8’s topic-block analysis interpretable at all.

5.4 Graph 4: Real versus shuffled correlations

Every analysis below runs on the same 91-respondent, ordinal-imputed matrix, so no part of this report silently uses a different sample.

Figure 5 compares the 1,770 observed item–item correlations against a null that destroys inter-item association while preserving every item’s marginal exactly. Each of 200 replicates permutes every column independently and re-runs the full pipeline — row-centre, then correlate — on the shuffled matrix. This detail is not cosmetic: one permutation applied to all columns, or permuting rows, merely relabels respondents and leaves the correlation structure numerically untouched.

Findings. Correlations from $n = 91$ respondents scatter about zero with SD $1/\sqrt{n-3} = 0.107$ even when nothing is related. The permuted null reproduces this at **0.106**, against **0.139** observed. Real structure exists but is modest, and lives in the tails:

$ r >$	0.15	0.20	0.25	0.30
Observed edges	502	266	124	51
Expected by chance	286	107	32	7
Chance fraction	57%	40%	26%	14%

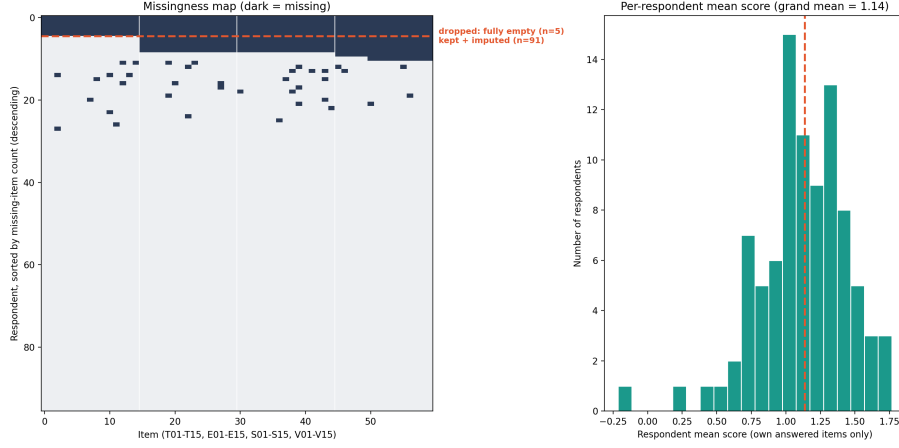


Figure 2: Missingness map and per-responder response means. Left: missingness over all 96 raw respondents, sorted by missing-item count. The dashed line marks the exclusion rule — only the five entirely empty records are dropped, and the remaining 91 have their 242 missing cells imputed. Right: per-responder mean over answered items, grand mean 1.14.

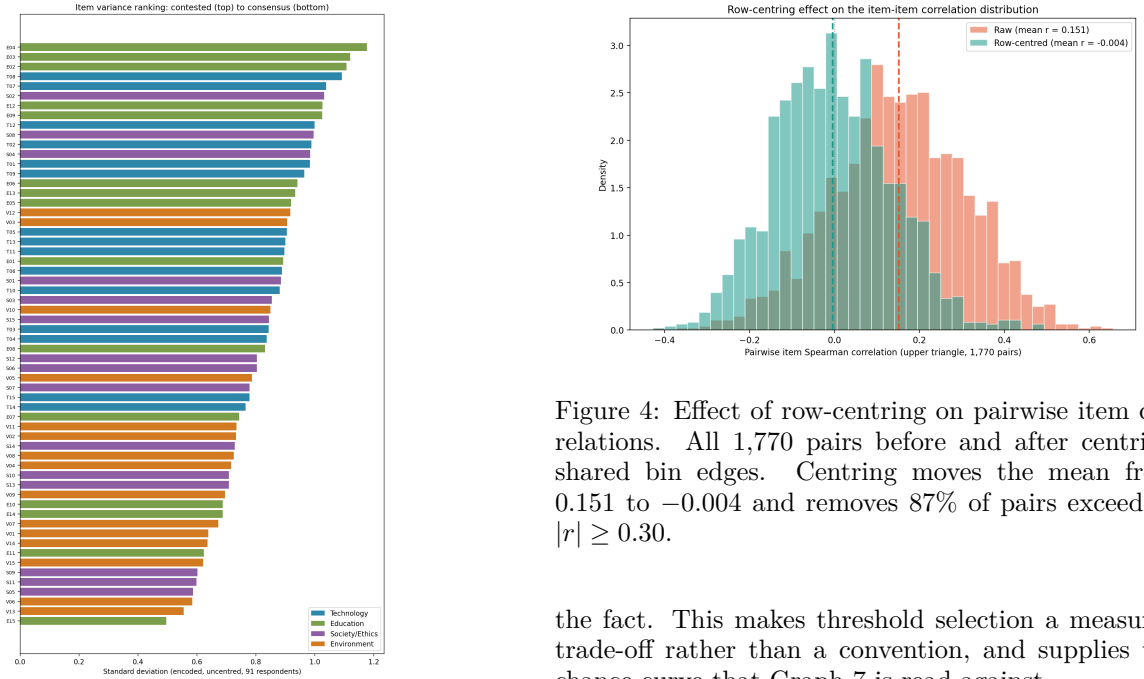


Figure 4: Effect of row-centring on pairwise item correlations. All 1,770 pairs before and after centring, shared bin edges. Centring moves the mean from 0.151 to -0.004 and removes 87% of pairs exceeding $|r| \geq 0.30$.

Figure 3: Item variance across the four survey blocks. All 60 items by standard deviation on the uncentred matrix, coloured by block. Contested items concentrate in education and technology; environment clusters at the consensus end.

Implication. At the conventional cutoff $|r| > 0.15$ more than half the surviving edges are what 1,770 simultaneous estimates from 91 people produce by luck. A network drawn there is mostly noise, and no downstream community detection can separate the two after

the fact. This makes threshold selection a measured trade-off rather than a convention, and supplies the chance curve that Graph 7 is read against.

5.5 Graph 5: How many dimensions are real?

Graph 4 shows a noise floor exists, but not how much of the matrix survives it. A plain PCA cannot answer this either: the first component explains only 8.3% of variance, which is just as consistent with weak real structure as with none. Marchenko–Pastur theory gives an exact answer instead: for a correlation matrix built from pure noise, the eigenvalues fall inside a known band, $\lambda_+ = (1 + 1/\sqrt{Q})^2$ with $Q = T/N$; anything above that band is structure T respondents

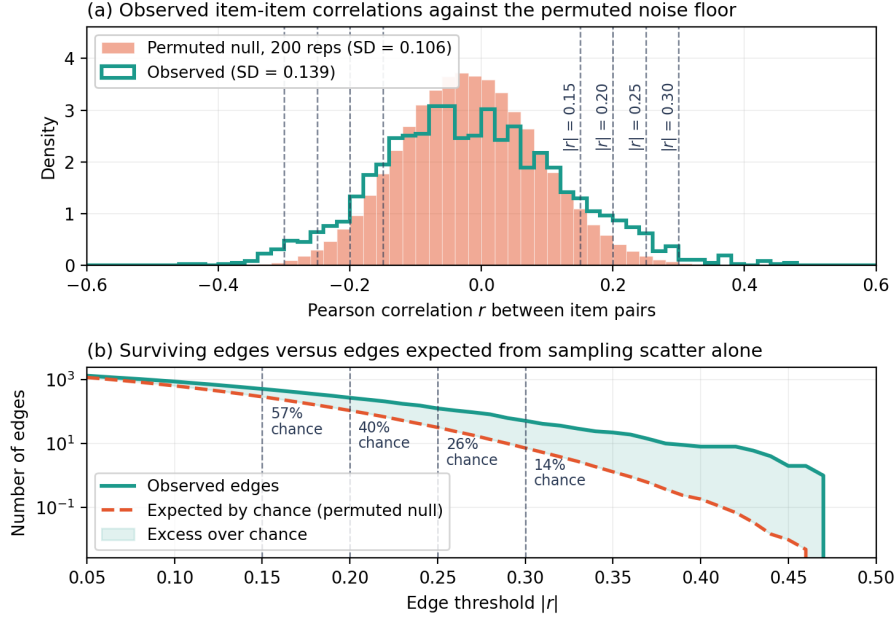


Figure 5: Observed versus permuted item correlations. (a) Observed correlations against a 200-replicate permutation null that re-runs the whole pipeline on independently shuffled columns; SD 0.139 against 0.106, the excess confined to the tails. (b) Surviving edges against edges expected from sampling scatter alone (log scale); the shaded band is the excess over chance.

could not produce by chance alone.

Findings. With 91 respondents and 60 items, the band’s upper edge is $\lambda_+ = \mathbf{3.283}$. Exactly four eigenvalues clear it — 4.38, 4.06, 3.55, 3.35 — carrying **25.6%** of the total variance, while the fifth, at 2.86, sits clearly inside the band (Figure 6). Two of the four modes align with the survey’s own topic blocks; the other two cut across them, pointing to attitudes that do not respect the four-topic design.

Implication. The remaining three quarters of the matrix is sampling noise. Every network built later in this report uses only these four modes, not the raw matrix, so it reflects only what 91 respondents can actually support.

5.6 Graph 6: Reordered correlation heatmap

Figure 7 presents the MP-denoised correlation matrix reordered by the community assignments detected in Graph 9 (Section 5.9). A diverging colour map centred at zero distinguishes positive from negative correlations, and the marginal colour strips mark the predefined thematic blocks (technology, education, society/ethics, environment).

Findings. The block-diagonal structure of the reordered matrix confirms that the community detection groups densely connected items. The marginal strips, however, show those data-driven communities cutting directly across the survey’s predefined thematic blocks,

and the formal comparison yields a normalised mutual information (NMI) of 0.205.

Implication. Respondents’ association patterns do not partition neatly into the four hypothesised domains. The partition is a good description of the graph, but it is not the partition the instrument was built around.

5.7 Graph 7: Choosing the correlation threshold

Picking a single threshold invites the objection that it was chosen to produce the result. Figure 8 answers this by rebuilding the network at all 46 cutoffs from 0.05 to 0.50 and tracking edge count, the giant component, fragmentation and modularity at each one, so the chosen value is visibly one point on a curve rather than a cherry-pick.

Findings. Two things pull in opposite directions as the threshold rises: the share of chance edges from Graph 4 falls, but the graph eventually breaks apart — by $|r| > 0.27$ the giant component covers under half the items. We pick the highest cutoff at which the giant component still spans two thirds of the items and mean degree stays above 4: this selects $|r| > \mathbf{0.22}$, giving **131** edges (68 positive, 63 negative), mean degree 4.37, and a giant component of 50 of 60 items (83%). At the next step, $|r| > 0.23$, the giant component no longer qualifies. An independent estimate from Graph 4’s chance-edge count agrees closely: about 138 real

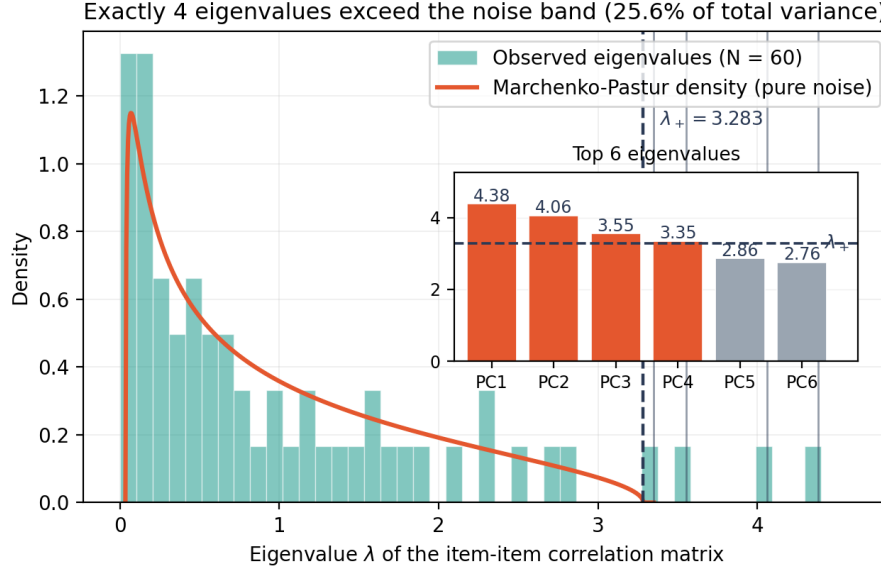


Figure 6: Eigenvalue spectrum against the Marchenko–Pastur noise band. The spectrum of the row-centred item correlation matrix against the Marchenko–Pastur density for pure noise of the same shape ($T = 91$, $N = 60$). Exactly four eigenvalues exceed $\lambda_+ = 3.283$, carrying 25.6% of variance. Inset: a clear gap between the fourth (3.35) and fifth (2.86).

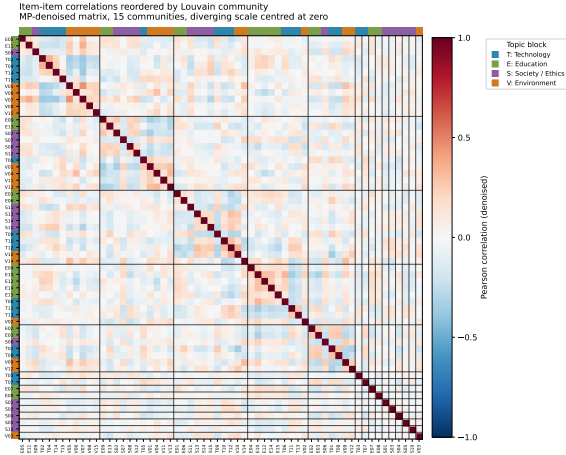


Figure 7: MP-denoised item correlation matrix reordered by community. The 15 communities are separated by black lines, with topic blocks marked in the margins. The communities cut across the predefined topic blocks, showing that respondents’ association patterns do not follow the survey’s own design.

edges expected there, against 131 kept here.

Implication. Modularity keeps climbing past this point, reaching 0.72 — but only once the graph has broken into 43 disconnected pieces. Quoting that number alone, without the rest of the sweep, would mistake fragmentation for community structure, the exact trap

Section 6.2 unpacks.

5.8 Graph 8: Topic-block connectivity

The questionnaire arrives with its own partition — fifteen items each for technology, education, society and environment — which is an editorial claim about what belongs together. Whether it is also a statistical one should be asked before any community result is interpreted against it. We collapse the 60×60 correlation matrix into a 4×4 block matrix, averaging the 105 distinct within-block pairs on the diagonal (self-correlations excluded: keeping the unit diagonal would manufacture the very block structure being tested). Significance comes from a block-label permutation — the matrix is held fixed and the T/E/S/V labels reshuffled over 5,000 seeded draws — which needs no threshold and is immune to any constant offset in the matrix.

Findings. Row-centring is what makes the design visible at all. On the raw matrix (Figure 9, left) all sixteen cells are positive (+0.041 to +0.377): acquiescence makes every topic look connected to every other. After centring, the sign pattern separates cleanly — every diagonal cell positive, every off-diagonal negative. Within-block mean $r = +0.0746$ against between-block -0.0282 gives a gap of **+0.1027** against a null of -0.00003 ± 0.0077 : $z = 13.4$, with no permutation in 5,000 reaching the observed value ($p < 0.0002$).

The blocks are not equally coherent: environment +0.136, technology +0.066, education +0.054, society/ethics

Threshold sweep on the MP-denoised item network

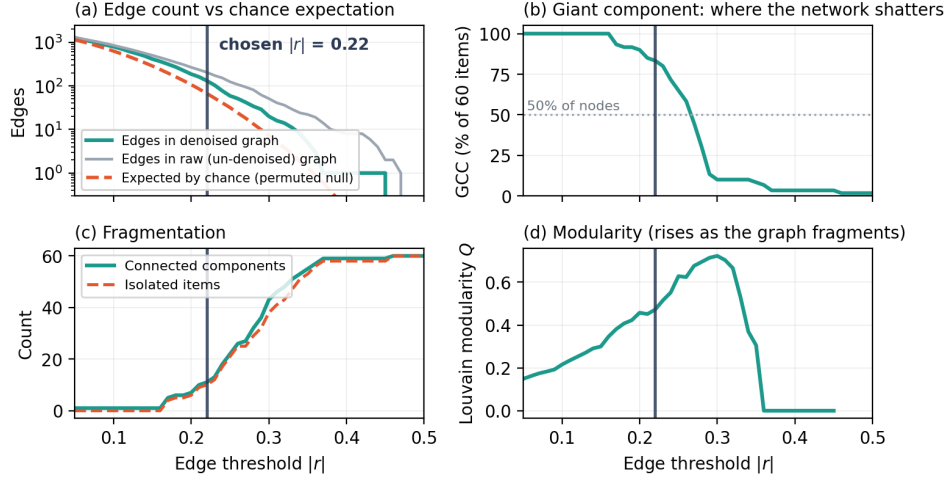


Figure 8: Threshold sweep and network fragmentation. The signed item network rebuilt at all 46 cutoffs from 0.05 to 0.50. (a) Edge count against the chance expectation from Graph 4. (b) Giant component. (c) Components and isolates. (d) Louvain modularity, which reaches 0.72 only once the graph has broken into 43 pieces. The vertical line marks the selected $|r| > 0.22$.

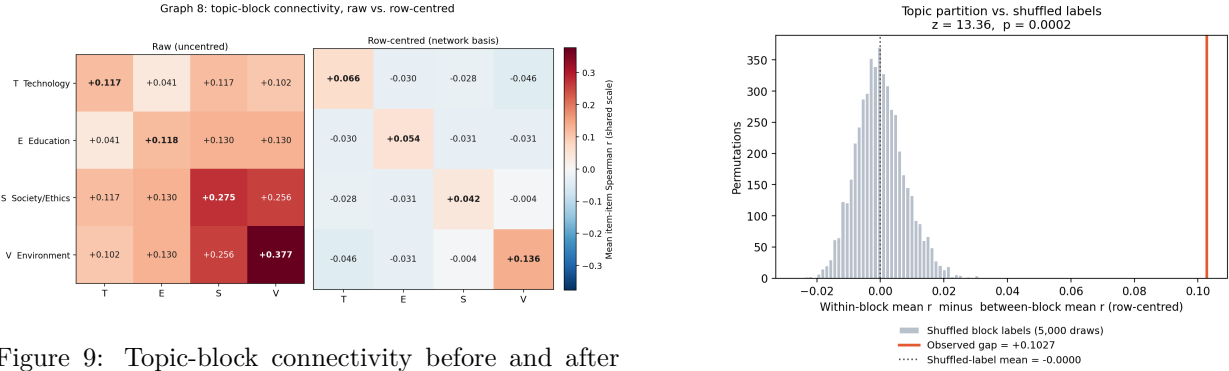


Figure 9: Topic-block connectivity before and after row-centring. Mean item-item correlation within and between blocks. Left: the raw matrix — every cell positive, the design invisible under acquiescence. Right: the same collapse after row-centring, shared colour scale; diagonal positive, all off-diagonals negative.

+0.042. Environment is more than three times as internally consistent as ethics. S-V is the one seam, at -0.004 the only pair not clearly separated and the strongest off-diagonal cell in the raw matrix (+0.256).

Implication. The thematic design is statistically real, so a later community that aligns with a block is recovering something the instrument built in, while one that cuts across blocks is the more informative object — which is exactly what Graph 9 finds (NMI 0.205). The uneven diagonal sharpens this: environmental attitudes travel as a package, whereas ethics

Figure 10: Is the block design statistically real? Distribution of the within-minus-between correlation gap under 5,000 reshufflings of the block labels. The observed gap, +0.1027 ($z = 13.4$), is nowhere near this shuffled-label distribution.

items are close to independent, so respondents hold ethical positions issue by issue rather than as a doctrine. One caveat: the negative off-diagonals do not mean topics oppose one another. Row-centring forces each row to sum to zero, pushing the grand mean of all 1,770 correlations to -0.0038 ; the values should be read relative to the diagonal, not as opposition.

5.9 Graph 9: Are the communities real?

Figure 11 compares three community-detection algorithms on the item network at $|r| > 0.22$. Figure 12 then checks the communities found against three random comparisons — degree-preserving rewiring, an Erdős–Rényi graph, and a column-permutation baseline that scrambles responses independently per item — to see whether the same result could appear by chance.

Which algorithm. Louvain found 15 communities at $Q = 0.472$, but its greedy merging step can leave a community internally disconnected. Leiden adds a fix for exactly this and finds a better partition at the same size, $Q = 0.491$. Girvan–Newman, an unrelated divisive method, independently peaks at $Q = 0.447$, also at $k = 15$. All three agree on 15 communities, so that count is a property of the graph rather than of one algorithm; Leiden is used from here on.

But the comparisons disagree. Leiden’s $Q = 0.491$ comfortably beats the Erdős–Rényi comparison (0.420 ± 0.016) and the rewiring comparison (0.333 ± 0.013). Against the column-permutation comparison, though, it does not just fail — it scores lower: 0.629 ± 0.044 against our 0.491. Data with the responses scrambled, pushed through the exact same pipeline, produces more modular graphs than the real data does. Section 6.2 explains why: the first two comparisons only reshuffle the finished graph, so neither can catch structure the graph-building process itself creates; only the column-permutation comparison rebuilds that process from scratch.

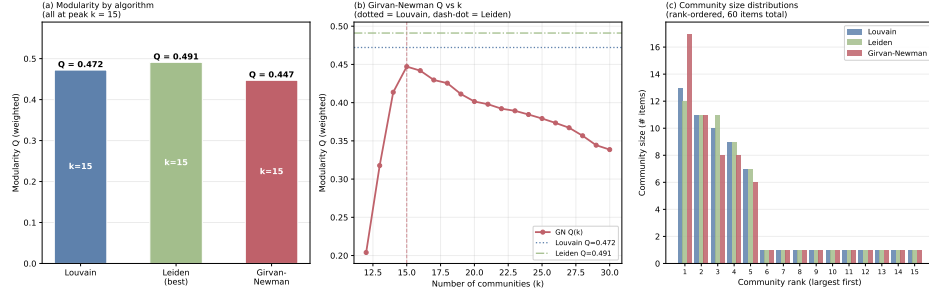


Figure 11: Community detection algorithms compared on the item network. (a) Peak modularity per algorithm; all converge on $k = 15$, Leiden highest at $Q = 0.491$. (b) The Girvan–Newman $Q(k)$ curve rising to a clear peak at $k = 15$ before declining, with Louvain and Leiden peaks marked. (c) Rank-ordered community sizes, similar across all three methods.

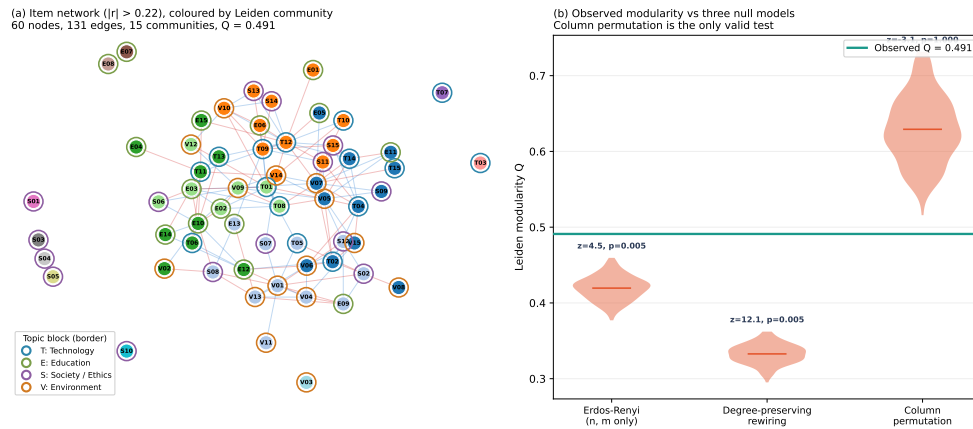


Figure 12: Item-network community structure against three comparisons. (a) The item network at $|r| > 0.22$, nodes coloured by Leiden community, borders marking the predefined thematic block. (b) Observed modularity $Q = 0.491$ sits comfortably above the Erdős–Rényi and degree-preserving rewiring comparisons but below the column-permutation comparison that reproduces the construction.

6 Results and Discussion

6.1 What the data supports

Superficial consensus, genuine local structure. The headline is not polarisation. Agreement runs at 79.8% across every block and PC1 explains about 8% of variance, so there are no two opposed camps. There is a broad agreeable consensus with real disagreement concentrated in identifiable places.

Disagreement is domain-specific, not dispositional. Environment and society are settled; technology and education are not. Within education the cohort is precise about what it rejects — compulsory attendance and written examinations — while endorsing curriculum reform above 94%. Within technology the objection tracks delegated consequential judgment (diagnosis, autonomous vehicles) rather than automation as such: robots in hazardous work still score 82.4%.

Most apparent structure was response style. Row-centring removes 87% of the item pairs exceeding $|r| \geq 0.30$. Any analysis skipping this step finds dense, uniformly positive structure and is measuring agreeableness.

Communities ignore the survey’s own design. The thematic blocks are statistically real ($z = 13.4$ on the within-minus-between gap, Graph 8), yet the 15 detected communities align with them only at $NMI = 0.205$. Respondents associate ideas across topic boundaries, not within the categories the questionnaire was built from.

6.2 What failed, and what that tells us

Three results did not come out the way the analysis was set up to expect. They are reported because each one changes what may legitimately be claimed.

1. Most of the correlation matrix is noise. Marchenko–Pastur filtering (Section 5.5) leaves exactly four eigenvalue modes above the band, carrying 25.6% of total variance. The other 74.4% is what 91 respondents answering 60 items generate by chance. Any claim resting on a single correlation, or on fine structure of the full matrix, is unsupported — which is why every downstream network is built from the denoised matrix rather than the raw one.

2. The community structure does not survive its own null. This is the sharpest negative result, and which null is used decides it. A correlation-derived network has transitivity built into its construction: if A correlates with B and B with C, A and C are pulled toward correlation automatically, and that alone manufactures modularity in structureless data. Rewiring and Erdős–Rényi nulls take the finished graph as given, so neither can detect structure the construction itself created. Only a column-permutation null — destroying association in the raw responses, then regenerating the entire chain of centre, correlate, threshold, measure — puts the claim actually being made under test. The three nulls disagree sharply:

Null model	Q_{null}	z
Degree-preserving rewiring	$0.333 \pm .013$	12.13
Erdős–Rényi	$0.420 \pm .016$	4.46
Column permutation	$0.629 \pm .044$	−3.14

Against the graph-level nulls the observed $Q = 0.491$ looks overwhelming, at z up to 12. Against the null that reproduces the construction it inverts: scrambled data pushed through the identical pipeline yields more modular graphs than the real data. Had we stopped at the conventional rewiring null we would have reported a strong, entirely spurious finding. The communities are real as a description of the graph; they are not evidence about the cohort.

3. The dimension count depends on who you keep. Filling the same 242 cells by column mean, column median or ordinal regression all yield exactly four modes (24.7%, 25.0%, 25.6% of variance) — the structure is a property of the data, not of the imputer. But dropping the ten respondents with more than twelve missing items and filling the rest with a column statistic gives only three modes at 19.3%. The count is robust to how missing cells are filled and fragile to which respondents are retained. Keeping all 91 is the better-supported choice — discarding usable respondents to avoid modelling 4.4% of cells trades away more than it protects — but the result is conditional on it.

7 Conclusion

We built an end-to-end pipeline from a raw ordinal questionnaire to a signed item network, measuring each intervention rather than asserting it. Ordinal-regression recovery of the 242 missing cells beats a modal baseline under both masking mechanisms, and row-centring eliminates 87% of the item pairs correlated at $|r| \geq 0.30$. A permutation null puts the noise floor at 0.106 against an observed spread of 0.139, Marchenko–Pastur filtering leaves four significant modes carrying 25.6% of variance, and the threshold came from a 46-point sweep by a stated rule at $|r| > 0.22$.

On the resulting network, the questionnaire’s own thematic design proves statistically real ($z = 13.4$), yet the 15 detected communities nonetheless cut across the four topic blocks (NMI 0.205).

The same framework produced the most useful negative result. That community structure clears the rewiring and Erdős–Rényi nulls but fails the column-permutation null — a finding we would have missed entirely had we used only the conventional null, and the clearest argument in this report for choosing a null that reproduces the construction. What survives is a cohort in superficial consensus with genuine but local structure: four modest cross-cutting dimensions and communities that ignore the survey’s own thematic design. Not two camps.

8 Individual Contribution

The work split into three roughly equal parts, one per member, and each member wrote up their own analyses in the report.

Member	Contribution
Arijeet Paul	Cleaned and encoded the raw responses, separating dropout from withheld answers; built and validated the model that recovers the missing cells; ran the row-centring that removes each respondent’s response-style bias. Wrote up Graphs 1, 2, 3 and 8: the missingness pattern, item-level disagreement, the effect of centring, and whether the survey’s topic blocks hold together statistically.
Shreyash Lohare	Designed the test for how much of the correlation matrix is genuine signal rather than sampling noise, applied Marchenko–Pastur filtering to find the four significant modes, and set the threshold used to build the network. Wrote up Graphs 4, 5 and 7: the noise floor, the eigenvalue spectrum, and the threshold sweep.
Dev Patel	Implemented and compared three community-detection algorithms, tested the resulting communities against three randomised comparisons for statistical significance, and produced the reordered correlation heatmap. Wrote up Graphs 6 and 9: the community structure and how it was tested.

All three members jointly wrote the Dataset Documentation, Pipeline Followed, Results and Discussion, and Conclusion sections, and reviewed each other’s analyses before submission.